

NATIONAL INSTITUTE OF TECHNOLOGY , AGARTALA



ELECTRONICS AND INSTRUMENTATION ENGINEERING

ELECTRICAL & ELECTRONIC INSTRUMENTATION AND MEASUREMENT LAB

**TITLE - Industrial gas measurement sensor
system.**

GROUP - 03

ANURADHA KUMARI - 23UEI154

MANJESH KUMAR - 23UEI144

ANJALI PATEL - 23UEI148

RAHUL GOSWAMI - 23UEI159

KRISHNA NETHAVATH - 23UEI150

SAUMYA SHREYA - 23UEI157



APPENDIX

1.Introduction

2.Objective

3.Components

4.Block Diagram

5.Circuit Diagram

6. About MQ-2/MQ-3/MQ-4/MQ-5/MQ-6/MQ-8 Sensor

7.Work Flow

8.Program

9.Reference

INTRODUCTION

Industrial environments often involve gases like methane (CH_4), carbon monoxide (CO), hydrogen (H_2), ammonia (NH_3), and LPG, which can be toxic, flammable, or harmful to workers and equipment. Monitoring these gases in real-time is essential for safety, environmental compliance, and process control.

This project aims to design and implement a gas measurement system that senses, measures, and displays the concentration of hazardous gases.

Gas sensors like MQ-5 / MQ-135 convert gas concentration into electrical signals that can be measured.

This project develops a low-cost, real-time gas measurement system with alarm functionality.

OBJECTIVE

- To design a system that can detect and measure harmful gases.
- To provide real-time indication of gas levels.
- To trigger an alarm when gas concentration exceeds safe threshold.
- To ensure worker safety and accident prevention in industrial environments.
- To demonstrate practical application of sensors and instrumentation principles.

COMPONENTS

Core Sensor

MQ-2,MQ-3,MQ-4,MQ-6,MQ-8

Processing

ArduinoUno

Output

16x2 LCDwith I2C module (display)

Buzzer for alarm

LEDs (Green = Safe, Red = Danger)

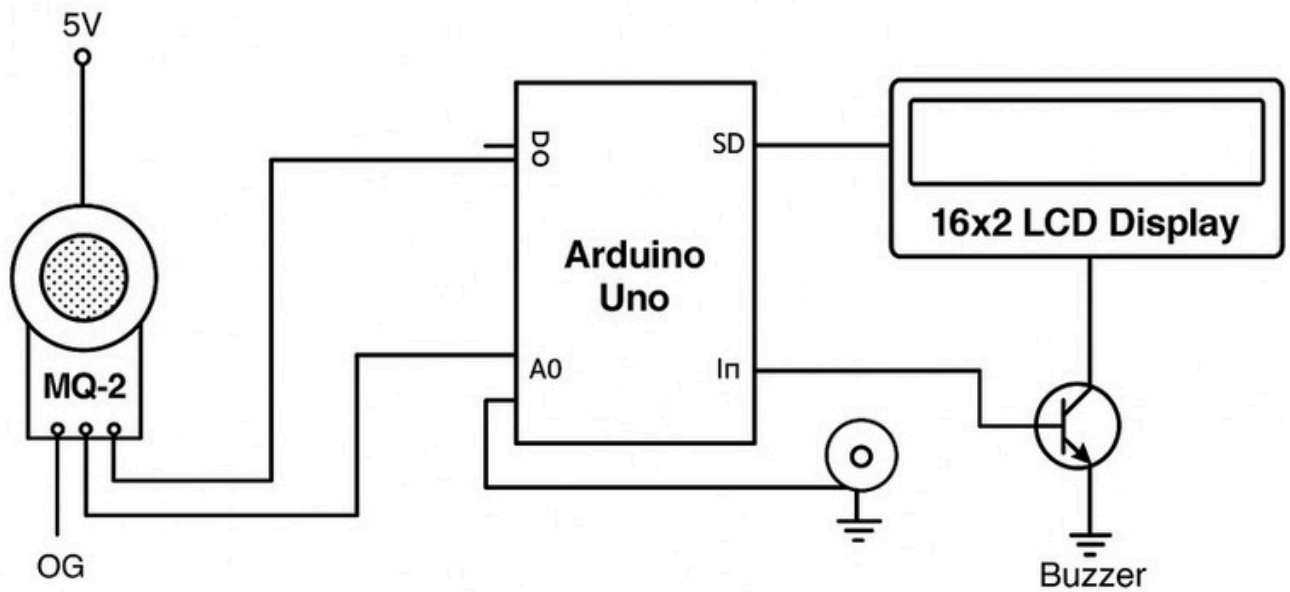
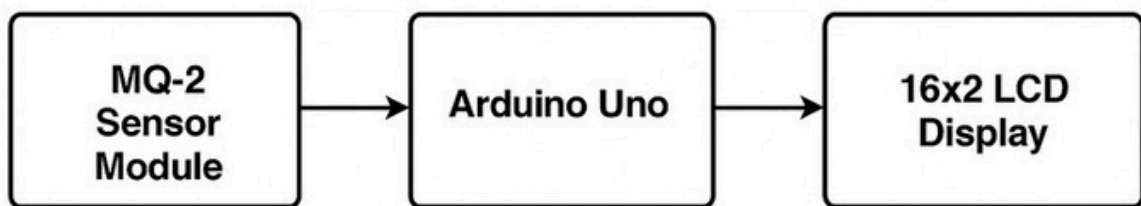
Power Supply

5VDCregulatedsupply

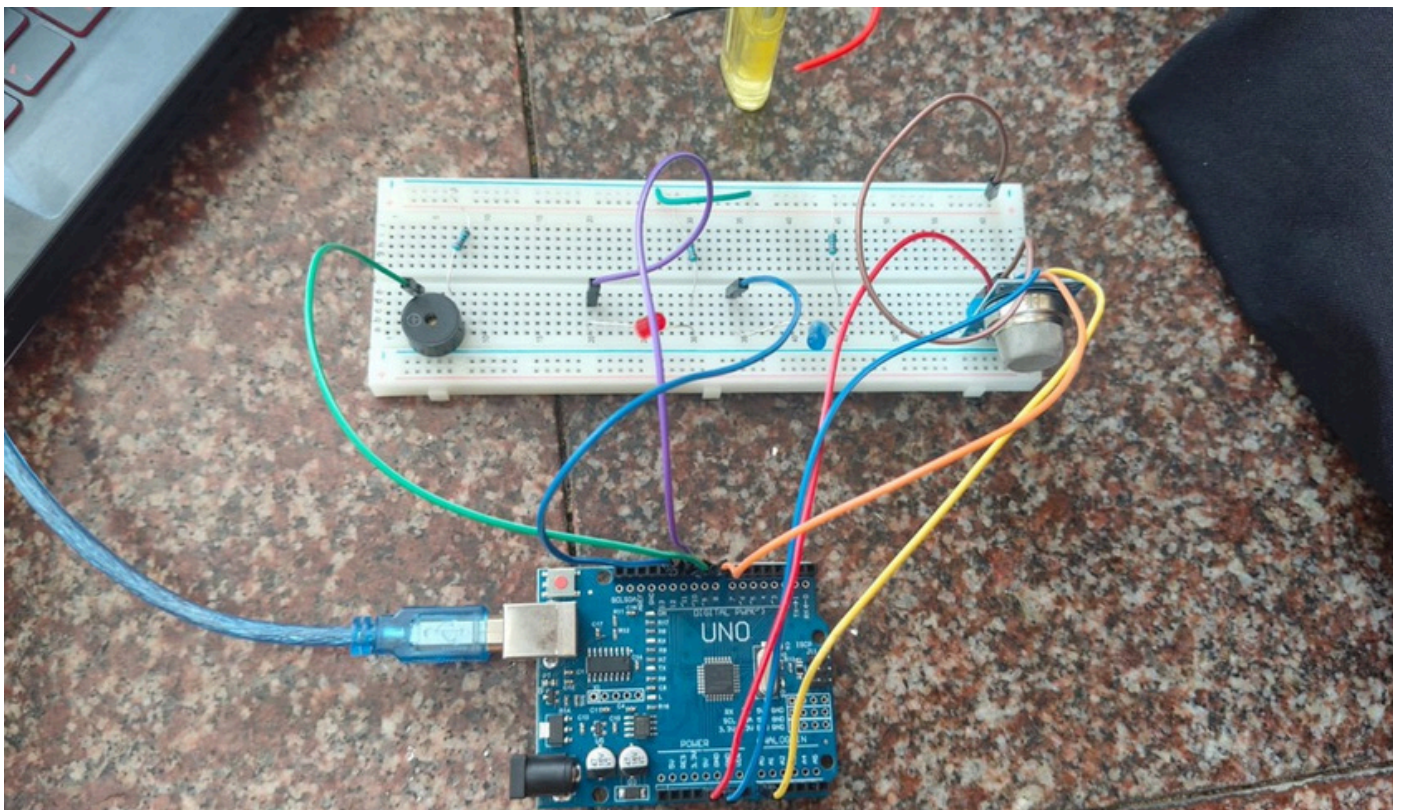
Miscellaneous

Resistors, capacitors,potentiometer, jumper wires,
breadboard.

BLOCK DIAGRAM



CIRCUIT DIAGRAM



MQ2 Gas Sensor

Introduction

The MQ-2 is a gas detection sensor commonly used to detect LPG (Liquefied Petroleum Gas), smoke, methane, propane, hydrogen, alcohol, and carbon monoxide.

It is one of the most popular sensors in electronic projects related to fire detection, gas leakage alarms, and air quality monitoring systems due to its low cost and high sensitivity

Working Principle

MQ-2 sensor works on the principle of change in resistance of the sensing material (SnO_2 - Tin Dioxide) when it comes into contact with specific gases.

- In clean air, the sensor's tin dioxide (SnO_2) has high resistance.
- When target gases (like LPG, methane, CO, etc.) are present, they react with oxygen on the sensor surface and reduce the resistance of SnO_2 .
- This change in resistance is measured as a change in output voltage, which is proportional to the gas concentration.

MQ-3 Gas Sensor

Introduction

The MQ-3 gas sensor is an alcohol detection sensor widely used to measure the concentration of alcohol vapors in the air or breath.

It is one of the most popular sensors used in breathalyzer circuits, vehicle safety systems, and alcohol detection projects.

The sensor is highly sensitive to alcohol vapors (ethanol) and less sensitive to benzene, smoke, and other gases, making it suitable for detecting blood alcohol concentration (BAC) through breath.

Working Principle

The MQ-3 sensor works on the semiconductor gas sensing principle. It uses Tin Dioxide (SnO_2) as the sensing material.

- In clean air, the sensor's resistance is high.
- When alcohol vapors are present, they get oxidized on the surface of SnO_2 when heated, causing a decrease in sensor resistance.
- This change in resistance is converted into a voltage output, which is proportional to the alcohol concentration in the air.

MQ-4 Gas Sensor

Introduction

The MQ-4 gas sensor is a methane (CH_4) gas detection sensor widely used for natural gas leakage detection in homes and industries.

It is also sensitive to LPG, butane, and hydrogen, making it ideal for use in gas leak alarms and safety monitoring systems.

The MQ-4 is a semiconductor-type gas sensor that operates on a 5V DC supply and provides both analog and digital outputs.

Working Principle

The MQ-4 sensor works on the principle of change in conductivity of tin dioxide (SnO_2) when exposed to combustible gases like methane.

- In clean air, SnO_2 has high electrical resistance.
- When methane or other combustible gases are present, they reduce oxygen adsorption on the surface and react with the sensor material, causing a drop in resistance.
- This change in resistance is measured as a voltage output, which corresponds to the gas concentration.

MQ-5 Gas Sensor

Introduction

The MQ-5 gas sensor is a semiconductor-type gas sensor designed to detect LPG (liquefied petroleum gas), natural gas (methane), and coal gas.

It can also detect hydrogen and alcohol vapors to some extent.

The MQ-5 is widely used in gas leakage detection systems and industrial safety applications because of its high sensitivity, low cost, and easy interface with microcontrollers like Arduino or ESP32.

Working Principle

The MQ-5 sensor works on the principle of variation in resistance of its sensing material, Tin Dioxide (SnO_2), in the presence of combustible gases.

- In clean air, oxygen molecules are adsorbed on the SnO_2 surface, creating a high resistance.
- When the sensor is exposed to LPG or methane, these gases react with the adsorbed oxygen and release electrons into the SnO_2 surface.
- This reaction reduces the sensor's resistance, and the change is converted into a voltage signal, which is proportional to the gas concentration.

MQ-6 Gas Sensor

Introduction

The MQ-6 gas sensor is a semiconductor-type sensor mainly used for detecting LPG (Liquefied Petroleum Gas), butane, and propane gases in the air.

It is highly sensitive to flammable gases, making it an ideal choice for gas leakage detection systems, industrial safety devices, and smart home safety applications.

The sensor is known for its quick response, low cost, and easy interfacing with microcontrollers such as Arduino, Raspberry Pi, or ESP32.

Working Principle

The MQ-6 sensor works on the principle that the electrical resistance of its sensing material (Tin Dioxide – SnO_2) changes when exposed to target gases.

- In clean air, oxygen molecules are adsorbed on the SnO_2 surface, leading to high resistance.
- When LPG, butane, or propane gases are present, they react with the adsorbed oxygen and release electrons, which decreases the resistance of SnO_2 .
- This change in resistance is measured as a voltage change, which corresponds to the gas concentration in the air.

MQ-8 Gas Sensor

Introduction

The MQ-8 gas sensor is a hydrogen (H_2) gas detection sensor that is highly sensitive to hydrogen and slightly sensitive to other flammable gases such as LPG, methane, and alcohol vapors.

It is widely used in hydrogen leakage detection systems, battery safety monitoring, and industrial applications where hydrogen gas is produced or used.

This sensor offers a fast response, stable performance, and low cost, making it suitable for safety and automation systems.

Working Principle

The MQ-8 sensor operates on the principle of change in electrical conductivity of Tin Dioxide (SnO_2) when exposed to hydrogen gas.

- In clean air, oxygen molecules are adsorbed on the surface of SnO_2 , creating a high resistance layer.
- When hydrogen gas comes into contact with the sensor, it reacts with the adsorbed oxygen and reduces the surface resistance by releasing electrons.
- This change in resistance is converted into an output voltage, which is proportional to the hydrogen gas concentration in the air.

Working Flow

- Gas sensors contain a semiconducting material (SnO_2).
- In presence of target gas, the resistance of the sensor changes.
- This resistance variation is converted into a voltage signal.
- Signal Conditioning: Filtered and amplified using Op-Amp.

Processing:

- Arduino converts analog signal $\rightarrow$ ppm values.
- OR comparator circuit checks if gas $>$ threshold.

Output:

- LCD shows gas concentration.
- Buzzer/LED alerts when unsafe level is detected.

Signal Conditioning:

The analog signal is filtered and stabilized using resistors and capacitors. Optionally, an Op-Amp can amplify or compare the signal to a reference threshold for faster alarm response.

Processing by Arduino: Arduino Uno's ADC (Analog-to-Digital Converter) reads the voltage signal. Using a calibration curve (R_s/R_o vs ppm from datasheet), Arduino converts it to gas concentration in ppm.

Code logic:

Read analog value. Compare with threshold value. Trigger outputs if limit exceeded.

Output Indications:

LCD Display (via I²C) shows real-time gas concentration (e.g., “CO: 75 ppm”). LEDs:

Green → Normal/Safe Red →
Dangerous (above threshold)

Buzzer: Activated when unsafe concentration is detected.

Alarm & Safety:

Arduino continuously monitors the sensor. When concentration crosses a preset limit, the buzzer and red LED alert workers instantly.

REFERENCE

Google

Chatgpt

Bell

Youtube

THANK YOU